

HT-Advisor technician heat-treatment sheet

Fatigue objective - LPBF Inconel 718 - non-HIP route

Input conditions

Field	Value
Primary objective	fatigue
Alloy and process	LPBF Inconel 718
Initial material state	EOS-like LPBF, as-built
Build orientation	vertical
Surface condition	machined
Representative section size	thin coupon
HIP benchmark included	No
Maximum furnace temperature	1100 C
Maximum practical cycle time	20 h
Fatigue validation target	R = 0.1; Nf = 1,000,000 cycles

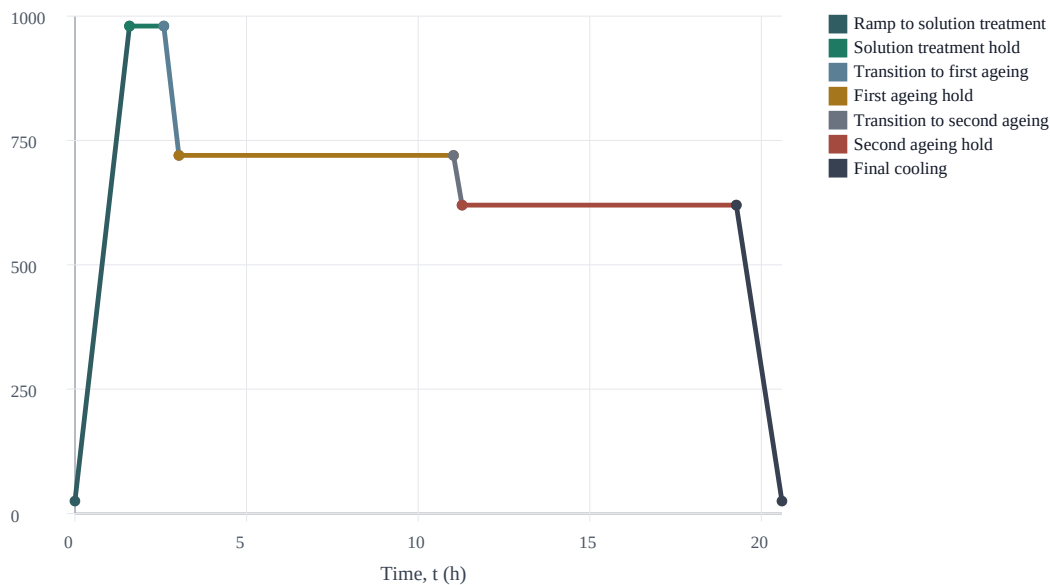
Recommended route

Field	Value
Route	ST_DA
Approved recipe	980 C for 1 h; 720 C for 8 h; 620 C for 8 h
Peak temperature	980 C
Total specified hold time	17.0 h
Recommendation index	0.77
Local feasibility	feasible under selected constraints

Nominal thermal programme

Step	Set point	Hold time	Action
1	980 C	1.0 h	Hold at temperature
2	720 C	8.0 h	Hold at temperature
3	620 C	8.0 h	Hold at temperature

Thermal step profile: temperature T (C)



The thermal profile is nominal. Record actual ramp rate, soak start and end times, cooling condition, specimen placement, and thermocouple or witness-coupon position.

Fatigue validation schedule

sigma_a (MPa)	R	sigma_max (MPa)	sigma_min (MPa)	Target runout
300	0.1	667	67	1000000
350	0.1	778	78	1000000
400	0.1	889	89	1000000
450	0.1	1000	100	1000000

Required process records

- Furnace ID: to be completed.
- Furnace programme ID: to be completed.
- Furnace atmosphere, vacuum, or shielding gas: to be completed.
- Thermocouple or witness coupon location: to be completed.
- Final cooling method: controlled furnace cooling unless locally approved otherwise.
- Operator sign-off and process-owner approval: to be completed.

Must-have experimental validation

- required: as-built baseline - Quantifies the starting LPBF Inconel 718 response before any thermal treatment.
- required: AMS-style standard baseline - Compares the framework-recommended route against a conventional solution plus double-ageing reference route.
- required: framework-recommended route: ST_DA - Tests the route selected by the decision-support workflow under the local furnace constraints.
- required: hardness and tensile testing - Validates the static property indicators used by the current metallurgy-informed model.
- required: SEM/EDS microstructural assessment - Checks Laves/Nb-rich segregation, precipitate response, and evidence that the selected heat treatment produced the intended metallurgical change.
- strongly recommended: fatigue or staircase screening if machine time is available - Non-HIP LPBF fatigue is defect-controlled; static tensile indicators alone do not establish fatigue safety.

S-N model boundary

Fatigue life is not predicted in the current release; stress ratio and target cycles are used only to plan validation tests.